

OCEANOGRAPHY

Crown-of-thorns starfish complete their larval phase eating only nitrogen-fixing *Trichodesmium* cyanobacteria

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Cyanobacteria of the genus *Trichodesmium* form extensive blooms that supply new N to nutrient-poor marine ecosystems. Yet little is known about what eats *Trichodesmium*. In this laboratory study, we show that one of the greatest threats to coral reefs, predatory crown-of-thorns starfish (CoTS), *Acanthaster* sp., completes their larval phase feeding solely on *Trichodesmium*. We observed *Trichodesmium* *erythraeum* CMP1985 in the stomachs of larvae using fluorescence microscopy and traced the assimilation of nitrogen from labeled trichomes into larval tissues using stable isotopes. Some larvae fed *T. erythraeum* were morphologically ready to become benthic juveniles after 19 days. Given that *Trichodesmium* can be food for CoTS, reported increases in *Trichodesmium* could be a driving factor in the heightened frequency of CoTS population irruptions that have devastated coral reefs in past decades. Future studies could test this through investigating the diets of wild larvae and incorporating *Trichodesmium* abundance into models of CoTS population dynamics.

INTRODUCTION

Global primary productivity relies on N₂ fixation by only a few types of cyanobacteria, one of which is the genus *Trichodesmium* (1, 2). Solitary multicellular trichomes and aggregations of trichomes are common throughout the year in tropical and subtropical oceans, where they migrate vertically throughout the photic zone (3, 4) but can also form seasonal surface blooms covering up to tens of thousands of square kilometers (5–7), with the timing of these blooms varying among regions (7, 8). The amount of nitrogen fixed by *Trichodesmium* spp. (hereafter *Trichodesmium*) is estimated to be 60 to 80 Tg of N per year, equivalent to 30 to 80% of global marine N₂ fixation (2, 9). Yet despite their prevalence and crucial role in N cycling, little is known about the trophic ecology of *Trichodesmium*. The few organisms known to prey on *Trichodesmium* are specialist copepods that live among aggregations (8, 10–12), although trichomes are ingested by four fishes, several salps, a pelagic crab, a penaeid prawn/shrimp, a pearl oyster, and one bird species (13–15).

At least 80% of all tropical marine benthic invertebrates spend time in the water column as larvae, consuming phytoplankton before settling to the benthos (16, 17). *Trichodesmium* are often the most abundant phytoplankton in tropical oceans (5, 18–20) but are overlooked as potential food for larvae; reputedly toxic, oversized, and nutritionally poor (2, 14, 15, 21, 22). The only study to feed *Trichodesmium* to the larvae of a non-pelagic marine invertebrate found larval banana prawns, *Penaeus merguensis*, ingested *Trichodesmium* sp. MACC0993 but did not develop beyond the first larval stage and displayed degeneration of gut cells consistent with starvation (14). Alternatively, it is possible that larvae use *Trichodesmium* as a source of nutrition by absorbing dissolved organic material (DOM) that leaks from trichomes (23, 24).

We report that larvae of the crown-of-thorns starfish (hereafter CoTS), *Acanthaster* sp., can complete their larval phase feeding solely on *Trichodesmium*. Adult CoTS are coral predators, capable of decimating Indo-Pacific coral reefs during population irruptions that contribute to coral reef ecosystem decline on Australia's world-heritage listed Great Barrier Reef (GBR) and elsewhere (25). Patterns in the ages of scars on long-lived massive corals, along with models of CoTS and coral population dynamics, suggest that population irruptions on the GBR during the past 50 years were more widespread and occurred more frequently than they did before the 1960s (26, 27). The causes of these “outbreaks” in CoTS numbers are hotly debated, but a leading hypothesis is that the normally low abundance of the larvae's phytoplankton food in oligotrophic tropical waters acts as a population bottleneck (25, 28). A heightened frequency of CoTS outbreaks has been attributed to anthropogenic terrestrial nutrient influxes that cause phytoplankton blooms and subsequent pulses of CoTS recruitment (28). Our results demonstrate that *Trichodesmium* can be a food source for CoTS that allows them to reach the juvenile stage. On the basis of this finding, we propose that reported increases in *Trichodesmium* availability in the GBR lagoon (18, 29, 30) could be a driving factor in the heightened frequency of CoTS outbreaks.

To show that CoTS are capable of ingesting *Trichodesmium*, in a laboratory feed trial, we fed 8-day-old CoTS larvae *Trichodesmium* *erythraeum* CMP1985 at three concentrations mimicking bloom and non-bloom conditions and observed trichomes in the stomachs of larvae after 3 and 24 hours. To demonstrate that larval CoTS digest *Trichodesmium*, we cultured *T. erythraeum* in medium enriched with isotopically labeled nitrogen (N) gas (¹⁵N-N) and measured increased ¹⁵N:¹⁴N ratios in both *T. erythraeum* and CoTS larvae fed ¹⁵N-labeled trichomes in a laboratory-scale feed trial. Last, to show that *Trichodesmium* supports normal larval development, we fed CoTS larvae *T. erythraeum* and DOM derived from *T. erythraeum* and followed development, growth, and survival in 1-liter cultures compared to unfed larvae and larvae fed *Proteomonas sulcata*, a cryptomonad microalga known to support rapid growth and development of larval CoTS (31). Some larvae fed *T. erythraeum* were morphologically ready to settle after 19 days, 6 days later than

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larvae fed *P. sulcata*. All larvae fed DOM and unfed larvae died by day 25.

RESULTS

Larvae ingest *Trichodesmium*

Eight-day-old larval CoTS exposed to *T. erythraeum* trichomes had material visible in the esophagus and stomach within a few hours (Fig. 1). The material in the digestive tract of larvae fed *T. erythraeum* fluoresced red when viewed under ultraviolet (UV) light at the wavelengths associated with fluorescence of chlorophyll (Fig. 1A). We also directly observed on multiple occasions CoTS larvae transfer a *T. erythraeum* trichome into their esophagus before rejecting the trichome or passing the trichome into their stomach (fig. S3). Unfed larvae did not have any fluorescent material in their digestive tract at any time.

After 3 hours, 28 to 78% of the CoTS larvae that were exposed to each of the two phytoplankton had material in their digestive tract (Fig. 1B). There were significantly more larvae with material in their digestive tract when fed *P. sulcata* than for larvae fed *T. erythraeum*, and this was independent of the density of phytoplankton [Fig. 1B; two-way analysis of variance (ANOVA) followed by pairwise test, table S1]. The proportion of larvae with material in their digestive tract when fed phytoplankton at a high density (1×10^4 cells ml^{-1} *P. sulcata* or 100 trichomes ml^{-1} *T. erythraeum*) was the same as for larvae fed phytoplankton at a medium density (1×10^3 cells ml^{-1} or 10 trichomes ml^{-1}) and significantly greater than when larvae were fed phytoplankton at a low density (1×10^2 cells ml^{-1} or 1 trichome ml^{-1}) (Fig. 1B and table S1). Unfed larvae did not have any material in their digestive tract.

After 24 hours, 31 to 96% of the larvae that were exposed to each of the two phytoplankton had material in their digestive tract

(Fig. 1C). There were significantly more larvae with material in their digestive tract when fed *P. sulcata* than for larvae fed *T. erythraeum*, and this was independent of the density of the phytoplankton (Fig. 1C; two-way ANOVA followed by pairwise test, table S1). The proportion of larvae with material in their digestive tract when fed phytoplankton at a high density was greater than for larvae fed phytoplankton at a low and medium density, and there were more larvae with food in their digestive tract when fed phytoplankton at a medium density than for those fed at a low phytoplankton density (Fig. 1C and table S1). Unfed larvae did not have any material in their digestive tract.

Larvae incorporate N fixed by *Trichodesmium*

Mean $\delta^{15}\text{N}$ {where $\delta^{15}\text{N} = [({}^{15}\text{N}/{}^{14}\text{N})_{\text{sample}}/({}^{15}\text{N}/{}^{14}\text{N})_{\text{air}}] - 1$ } of larval CoTS ranged from 7.5 to 164.8 per mil and depended on the feeding treatment that larvae were exposed to (Fig. 2; $F_{4,18} = 486.97$, $P < 0.0001$, followed by Tukey *b* test). Larvae fed ${}^{15}\text{N}$ -labeled trichomes (i.e., *T. erythraeum* grown in culture medium with ${}^{15}\text{N}$ - N_2 gas headspace) had significantly higher $\delta^{15}\text{N}$ than larvae fed *T. erythraeum* grown in standard culture medium (i.e., unlabeled trichomes) or unfed larvae (Fig. 2). When ${}^{15}\text{N}$ -labeled *T. erythraeum* culture was separated into solid (trichomes) and dissolved fractions (DOM) and given to larvae, the $\delta^{15}\text{N}$ of larvae fed labeled trichomes only was significantly greater than all other treatments, whereas the $\delta^{15}\text{N}$ of larvae fed labeled DOM did not differ from larvae fed unlabeled trichomes or unfed larvae (Fig. 2).

Larvae fed *Trichodesmium* develop normally

Growth and development of larval CoTS to 25 days postfertilization (dpf) was influenced by their diet (Figs. 3 and 4). At 10 dpf, larval diet treatment had a significant effect on the length ($F_{3,36} = 3.40$,

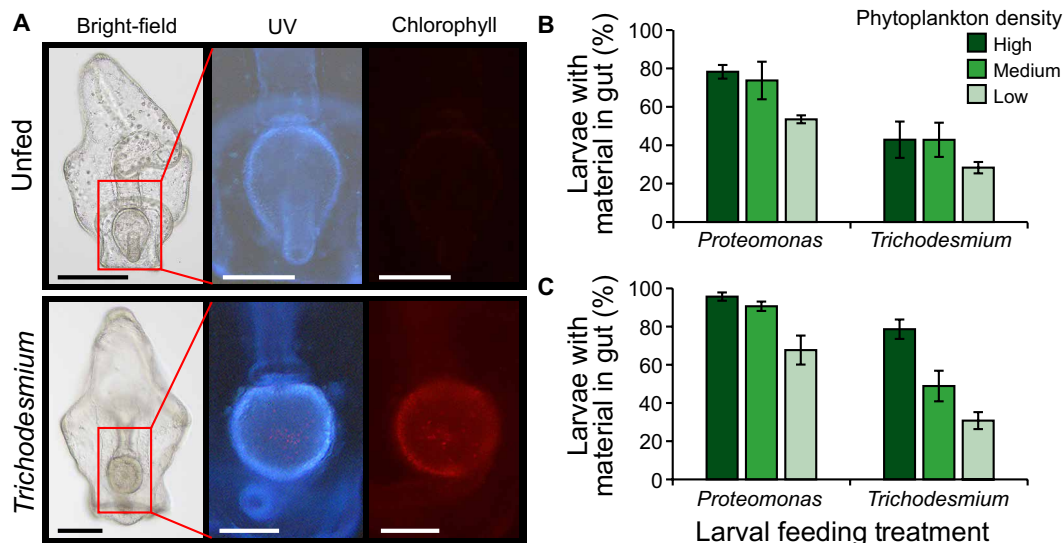


Fig. 1. Ingestion of filamentous cyanobacteria *T. erythraeum* CCMP1985 by 8-day-old CoTS (*Acanthaster* sp.) larvae. (A) Representative photographs of the digestive tract of CoTS larvae that were maintained in seawater without phytoplankton (Unfed) or seawater containing 100 trichomes ml^{-1} *T. erythraeum* for 3 hours, viewed using a compound microscope (40 $\times$ or 100 $\times$ zoom) under bright-field, unfiltered ultraviolet light (UV), and UV light with Olympus 19010 Chlorophyll Longpass filter (chlorophyll). Scale bars, 100 μm (black) and 50 μm (white). Original photos are in the Supplementary Materials (figs. S1 and S2). (B and C) Proportion of CoTS larvae that had fluorescing material in their digestive tract after (B) 3 hours and (C) 24 hours. Larvae were maintained in seawater containing a cryptomonad alga known to be readily consumed by CoTS, *P. sulcata* CS-412 (*Proteomonas*) or *T. erythraeum* CCMP1985 (*Trichodesmium*) at three densities: high, 1×10^4 cells ml^{-1} (*P. sulcata*) or 100 trichomes ml^{-1} (*T. erythraeum*); medium, 1×10^3 cells ml^{-1} or 10 trichomes ml^{-1} ; or low, 1×10^2 cells ml^{-1} or 1 trichome ml^{-1} . Larvae in an unfed control treatment did not have phytoplankton in their digestive tract at any time (not shown). Data are means $\pm$ SE, $n = 3$.

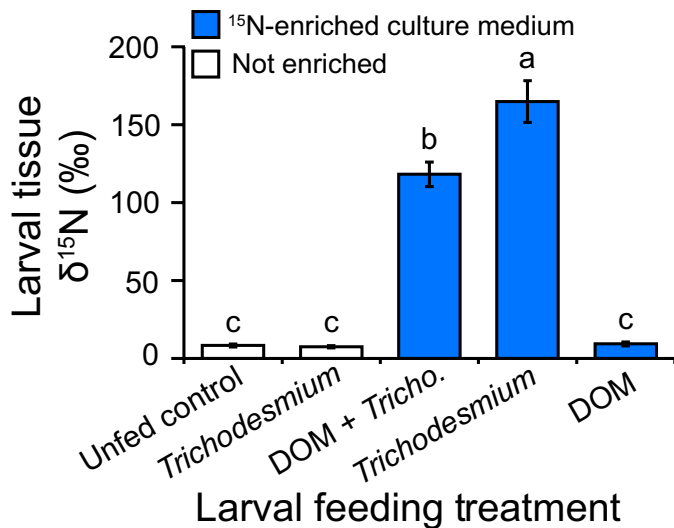


Fig. 2. $\delta^{15}\text{N}$ of larval CoTS, *Acanthaster* sp., fed *T. erythraeum* CCMP1985 and DOM derived from *T. erythraeum* cultures for 48 hours in a laboratory experiment. Before feeding, *T. erythraeum* was grown for 5 days in sealed bottles with an ambient air headspace (not enriched) or ^{15}N -N gas headspace (^{15}N -enriched culture medium). *T. erythraeum* and DOM were segregated via 0.45- μm filtration before feeding (see Materials and Methods). Bars that have a common letter are not significantly different according to ANOVA, $F_{4,18} = 486.97$, $P < 0.00001$, followed by Tukey b test. Data are means $\pm$ SD, $n = 4$ for not enriched treatments and $n = 5$ for ^{15}N -enriched culture medium treatments. ‰, per mil.

$P < 0.028$, followed by Tukey b test; Fig. 3A), width ($F_{3,36} = 3.25$, $P < 0.033$; Fig. 3A), and area ($F_{3,36} = 4.64$, $P < 0.008$; Fig. 3A) of larval CoTS but did not affect the length of ciliated bands ($F_{3,36} = 1.60$, $P = 0.206$; Fig. 3A). Larvae fed *P. sulcata* were longer than larvae that were not fed (Fig. 3A). There was no statistical difference in the length of larvae fed *P. sulcata*, *T. erythraeum*, and DOM (derived from *T. erythraeum*) (Fig. 3A) and no statistical difference in the length of larvae fed *T. erythraeum*, DOM, and unfed larvae (Fig. 3A). Larvae fed *P. sulcata* were wider than larvae that were not fed (Fig. 3A). There was no statistical difference in the width of larvae fed *P. sulcata*, *T. erythraeum*, and DOM (Fig. 3A) and no statistical difference in the width of larvae fed *T. erythraeum*, DOM, and unfed larvae (Fig. 3A). Larvae fed *P. sulcata* had greater area than larvae that were not fed and larvae fed *T. erythraeum*, which did not differ (Fig. 3A). The area of larvae fed DOM was not statistically different than all other treatments (Fig. 3A).

Growth of larvae fed *P. sulcata* was not followed beyond 10 dpf as larvae in this treatment reached their developmental endpoint (late brachiolaria) at 13 dpf (Fig. 4). At 16 dpf, larval diet treatment had a significant effect on the length ($F_{2,27} = 3.68$, $P < 0.038$, followed by Tukey b test; Fig. 3B), width ($F_{2,27} = 4.77$, $P < 0.017$; Fig. 3B), and length of ciliated bands ($F_{2,27} = 5.55$, $P < 0.010$; Fig. 3B) of larval CoTS but did not affect area ($F_{2,27} = 1.64$, $P = 0.214$; Fig. 3B). Larvae fed *T. erythraeum* were longer than larvae that were not fed (Fig. 3B). There was no statistical difference in the length of larvae fed *T. erythraeum* and DOM (Fig. 3B) and no statistical difference in the length of larvae fed DOM and unfed larvae (Fig. 3B). Larvae fed *T. erythraeum* were wider than larvae that were not fed (Fig. 3B). There was no statistical difference in the width of larvae fed *T. erythraeum* and DOM (Fig. 3B) and no statistical difference in the

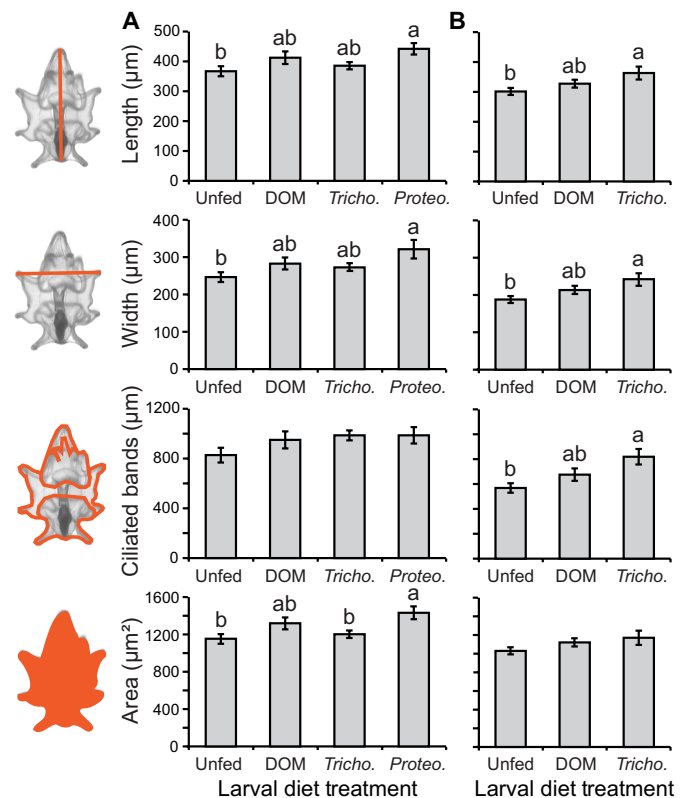


Fig. 3. Effect of diet on the morphology of larval CoTS (*Acanthaster* sp.). (A) Ten dpf (days postfertilization). (B) Sixteen dpf. Larvae were grown in 1-liter rearing containers and fed from 5 dpf with *T. erythraeum* CCMP1985 at a concentration of 20 trichomes ml^{-1} (*Tricho.*), DOM derived from *T. erythraeum* cultures (DOM), a cryptomonad alga known to support normal development of CoTS larvae, *P. sulcata* CS-412 at a density of 5×10^4 cells ml^{-1} (*Proteo.*), or were not fed (Unfed). The *P. sulcata* treatment was not followed beyond 13 dpf as some larvae in this treatment had reached their developmental endpoint (late brachiolaria) indicative of competency to metamorphose and settle into the juvenile lifestage (Fig. 4). Within each graph, bars that have a common letter are not significantly different according to ANOVA, $P < 0.05$ (in text), followed by Tukey b post hoc test. Data are means $\pm$ SE, $n = 10$.

width of larvae fed DOM and unfed larvae (Fig. 3B). Larvae fed *T. erythraeum* had longer ciliated bands than larvae that were not fed (Fig. 3B). There was no statistical difference in the length of the ciliated bands of larvae fed *T. erythraeum* and DOM (Fig. 3B) and no statistical difference in the width of larvae fed DOM and unfed larvae (Fig. 3B).

Development of CoTS larvae varied among larval feeding treatments (Fig. 4). At 13 dpf, there were no differences in the proportion of abnormal larvae among treatments ($F_{3,30} = 0.24$, $P = 0.870$; mean, 18.1 to 23.5%; Fig. 4A). However, larvae fed *P. sulcata* were substantially more developed compared to all other treatments, with 46.6% at the early brachiolaria stage, and some (~1 in 20) at the final larval lifestage, late brachiolaria (Fig. 4A). Some larvae fed *T. erythraeum* and DOM reached the early brachiolaria stage (5.4 and 6.3% respectively; Fig. 4A), but most (>70%) were bipinnaria. Unfed larvae had not developed beyond the bipinnaria stage (Fig. 4A). Development of larvae fed *P. sulcata* was not followed beyond 13 dpf.

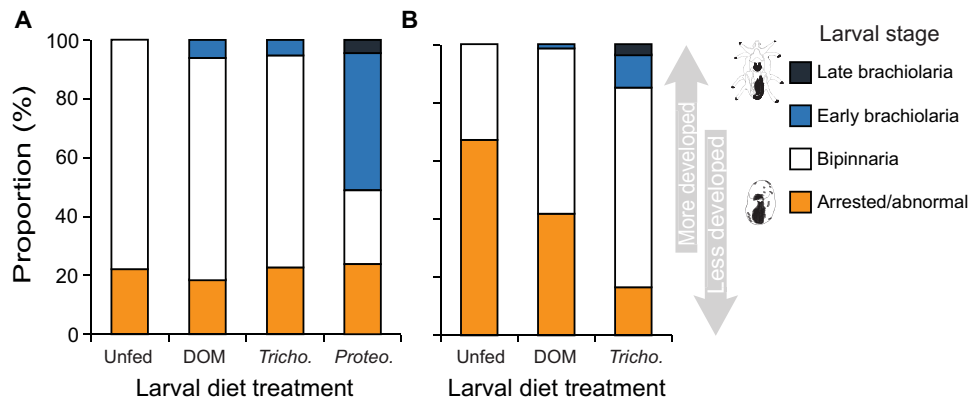


Fig. 4. Effect of diet on the development of larval CoTS (*Acanthaster* sp.). (A) Thirteen dpf. (B) Nineteen dpf. Larvae were grown in 1-liter rearing containers and fed from 5 dpf with *T. erythraeum* CCMP1985 at a concentration of 20 trichomes ml^{-1} (*Tricho.*), DOM derived from *T. erythraeum* cultures (DOM), a cryptomonad alga known to support rapid development of CoTS larvae, *P. sulcata* CS-412 at a density of 5×10^4 cells ml^{-1} (*Proteo.*), or were not fed (Unfed). The *P. sulcata* treatment was not followed beyond 13 dpf. Data are means $\pm$ SE, $n = 4$ to 10.

At 19 dpf, there were substantial differences in the proportion of abnormal larvae present in different larval diet treatments although ANOVA was unable to detect a significant difference ($F_{2,11} = 3.34$; $P = 0.074$; Fig. 4B). Most unfed larvae were abnormal (66.8%), a greater proportion than for larvae fed DOM (41.7%; Fig. 4B), which had more abnormal larvae present than larvae fed *T. erythraeum* (16.0%; Fig. 4B). Larvae fed *T. erythraeum* were more developed than larvae in other treatments, with 11.1% at the early brachiolaria stage and some (~1 in 25) at the late brachiolaria lifestage (Fig. 4B). Some larvae fed DOM were also at the early brachiolaria stage (1.4%; Fig. 4B), but the majority (56.9%) were bipinnaria. Unfed larvae were bipinnaria or abnormal (Fig. 4B).

Survival of larvae in different diet treatments decreased over time, with differences among treatments evident at some days (fig. S4 and table S2). For instance, survival at 13 dpf was significantly different among treatments (mean *T. erythraeum* = $22.3\% \pm 1.8$ SE; DOM = $17.8\% \pm 2.4$; unfed = $14.3\% \pm 2.9$; *P. sulcata* = $7.5\% \pm 2.3$; repeated measures ANOVA, table S2; *T. erythraeum* = DOM > DOM = unfed > unfed = *P. sulcata*). However, survival at 19 dpf was not different among treatments (mean *T. erythraeum* = $4.9\% \pm 1.1$ SE; DOM = $3.6\% \pm 1.4$; unfed = $3.2\% \pm 1.8$; survival of larvae fed *P. sulcata* was not followed beyond 13 dpf; table S2; *T. erythraeum* = DOM = unfed). Survival of larvae fed *T. erythraeum* was not followed beyond 19 dpf. At 22 dpf, there were larvae alive in 6 of the 10 replicates fed DOM, but only living larvae in 1 of the 10 unfed replicates. All remaining larvae in unfed and DOM treatments died by 25 dpf without developing to the late brachiolaria lifestage.

DISCUSSION

Larval CoTS ate *T. erythraeum* CCMP1985, adding to the short list of species known to ingest cyanobacteria of the genus *Trichodesmium* (8, 10–14). This is unexpected given that cyanobacteria are thought to be avoided by zooplankton (14, 21, 22) and previous studies showed larval CoTS to be highly selective feeders, preferring medium-sized phytoplankton (~1 to 20 μm) (32, 33). The filamentous shape of *T. erythraeum* trichomes did not prevent ingestion by larval CoTS, which ate trichomes from either distal end. It may be the diameter of trichomes, rather than length, that plays an

important role in determining their vulnerability to predation. Given that many larvae ingest particles that are substantially larger in diameter than *T. erythraeum* (34–36), *Trichodesmium* is likely ingested by a greater array of organisms than is presently recognized.

We traced the ^{15}N incorporated into *T. erythraeum* cells into the tissue of larval CoTS. This ^{15}N could only have been obtained via digestion of ^{15}N -labeled trichomes. Larvae were observed with whole and lysed *T. erythraeum* cells in their stomachs, and the $\delta^{15}\text{N}$ signature was greatest in larvae fed labeled trichomes without DOM. Echinoderm larvae have two or three types of stomach cells to digest phytoplankton via ingestion and intracellular digestion of whole phytoplankton cells or secretion of enzymes to digest polysaccharides in cell walls (37, 38). As these cell types likely originated from a common ancestor of deuterostomes (39), larvae with the ability to integrate N from *Trichodesmium* may be common among marine deuterostome taxa.

CoTS larvae fed *T. erythraeum* exclusively had similar survival rates to larvae fed a highly nutritious cryptomonad microalga and completed larval development in 19 days. This demonstrates that *Trichodesmium* provides sufficient nutrition to support normal growth and development of a benthic invertebrate, similar to pelagic copepods that live among *Trichodesmium*, which also prey on trichomes (8, 10–12). Larval banana prawns, *P. merguensis* fed *Trichodesmium* sp., and juvenile pearl oysters, *Pinctada maxima* fed *T. erythraeum*, appear to starve (14, 15). Juvenile white shrimp, *Penaeus vannamei*, die when exposed to a crude extract of mixed *Trichodesmium* (40). It is unclear why “innutritious” and “toxic” *Trichodesmium* can be food for CoTS. *T. erythraeum* is the most common *Trichodesmium* in the GBR (20), so CoTS from the GBR may be adapted to eat it. *T. erythraeum* may also not produce toxins (15, 22).

Larval CoTS fed *T. erythraeum* were on average smaller and took longer to reach morphological competence to settle compared to larvae fed the highly nutritious *P. sulcata* in this study (13 days) and elsewhere [16 to 19 days; (31, 41–43)]. Longer periods spent as larvae may increase exposure to predation (16, 44). Smaller larval size is also linked to poorer fitness in juveniles (45, 46). Nevertheless, 19-day pelagic larval duration (PLD) is within the norm for CoTS (31, 42), and PLD and larval morphologies are likely to be more variable in nature than in benign laboratory conditions. It is also likely that CoTS larvae fed varied densities of *Trichodesmium*

develop at different rates. We raised larvae feeding *T. erythraeum* at a rate of 20 trichomes ml^{-1} , a density associated with *Trichodesmium* blooms and >20 times more dense than typical abundances (5, 18, 19). High densities of eukaryotic phytoplankton mimicking blooms on the GBR slow growth of larval CoTS (41, 47). As *Trichodesmium* does not exist as monocultures in the wild (2, 19, 20), further research is warranted to identify how variations in *Trichodesmium* densities interact with other phytoplankton foods to determine the success of larval CoTS.

Trichodesmium may excrete half of newly fixed N, primarily as glutamate (23, 24). Lucas (47) posited that larval CoTS use DOM to obtain nutrients in oligotrophic ocean conditions. Subsequent work by Hoegh-Guldberg (48) demonstrated CoTS absorb DOM. It is not clear then why we found larvae exposed to DOM from ^{15}N -enriched *T. erythraeum* cultures had the same $\delta^{15}\text{N}$ levels as unfed larvae. Perhaps CoTS were unable to uptake or incorporate N in the forms that were present or were only using DOM as an energy source. Previous studies that have fed DOM to CoTS and other invertebrate larvae indicate that DOM is used as an energy source and if fed exclusively DOM may not provide sufficient nutrition for normal growth and development (48–50). Larval size is often correlated with energy reserves (45, 46), and CoTS larvae fed DOM in this study were larger than unfed larvae at the same age. CoTS fed DOM did not develop beyond the early brachiolaria stage and died by 25 dpf. It appears then that CoTS larvae primarily gain nutrients from *Trichodesmium* by eating trichomes, although uptake of DOM may still be important in providing energy when phytoplankton are scarce.

Given that CoTS larvae can complete their larval phase feeding on *Trichodesmium*, we raise the possibility that outbreaks of CoTS are linked to increased *Trichodesmium* abundances over the past century (18, 29, 30). The leading paradigm explaining population irruptions links mass CoTS recruitment events with increased abundances of eukaryotic microalgae associated with terrestrial runoff but has key flaws (25, 51). For instance, there is often poor correlation among floods/nutrient pollution, microalgae blooms, and CoTS population dynamics for many of the regions where population irruptions occur (51). Our results suggest that the success of CoTS larvae could be enhanced by increased *Trichodesmium* abundances irrespective of the abundances of eukaryotic microalgae that cannot fix N. The ability of CoTS larvae to feed on *Trichodesmium* to overcome oligotrophic conditions may also imply that recruitment by larval CoTS to the benthic juvenile stage may be more stable over multiple years than is presently known. A steady supply of small numbers of recruits combined with the ability of herbivorous CoTS to stall growth for at least 6.5 years (52, 53) could enable juvenile populations to build up to outbreak densities over time and make a synchronous transition to eating coral when conditions are optimal. Clearly, more work is needed to determine whether *Trichodesmium* plays a role in CoTS outbreaks. However, we have provided a crucial piece of evidence to support the idea that present-day and predicted future changes in the abundance of *Trichodesmium* and other cyanobacteria are likely to have far-reaching implications for the ecology of coral reef systems worldwide (2, 18, 54).

MATERIALS AND METHODS

Study species

Organisms used in this study are excluded from Australian state and federal ethics legislations, and, therefore, no ethics approvals were

required or sought. Adult *Acanthaster* sp. Gervais, 1841 were collected from the GBR near Cairns, QLD, Australia, in November 2021 and maintained in flow-through seawater (25° to 27°C) at the National Marine Science Centre in Coffs Harbour, NSW, Australia, until they were spawned in April 2022 using previously described methods (32, 42–44). Gametes from three males and three females were mixed, and resulting embryos were raised in 300-liter tanks until they were used in experiments following previously described methods (31, 41).

T. erythraeum CMP1985 was obtained from the National Center for Marine Algae and Microbiota (USA) and batch-cultured using a modified YBC-II medium under light at $\sim 80 \mu\text{mol m}^{-2} \text{s}^{-1}$ (LI-COR, USA) with a 22-hour:2-hour light:dark cycle at $\sim 25^\circ\text{C}$. The base of the medium was natural seawater obtained through the seawater intake system at the National Marine Science Centre, Southern Cross University (typical parameters: salinity of 34 to 36 and pH 8.0 to 8.05), filtered to $0.45 \mu\text{m}$ and autoclaved (121°C for 15 min), and allowed to sit for at least 3 days for gases to equilibrate, to which $\text{NaH}_2\text{PO}_4\cdot\text{H}_2\text{O}$ solution (6.9 g liter^{-1}), trace metal solution (mixed as per standard YBC-II medium recipe), and vitamin solution (mixed as per standard YBC-II medium recipe) were added at a rate of 1 ml per liter of seawater. *P. sulcata* CS-412 was obtained from the Australian National Algae Culture Collection [Commonwealth Scientific and Industrial Organization (CSIRO) ANACC, Hobart, AUS] and batch-cultured in aerated 17-liter carboys using F medium (CSIRO Modification) under light at $>150 \mu\text{mol m}^{-2} \text{s}^{-1}$ with a 22-hour:2-hour light:dark cycle at 25° to 31°C .

Larvae ingest *Trichodesmium*

CoTS larvae (5 dpf) were exposed to seven larval feeding treatments: *T. erythraeum* at three densities (high, 100 trichomes ml^{-1} ; medium, 10 trichomes ml^{-1} ; and low, 1 trichome ml^{-1}), *P. sulcata* at three densities (high, 1×10^4 cells ml^{-1} ; medium, 1×10^3 cells ml^{-1} ; and low, 1×10^2 cells ml^{-1}), and an unfed control (Unfed). Each treatment had three replicates, which consisted of a polypropylene vial (Techno Plas P8027UU) containing the larval feeding treatment and FSW ($0.22\text{-}\mu\text{m}$ filtered seawater) to a total volume of 30 ml, and 80 CoTS larvae that had not been fed during the preceding 48 hours. Vials were arranged on a benchtop at room temperature ($\sim 25^\circ\text{C}$) and a 22-hour:2-hour (light:dark) photoperiod. In all feeding trials, when observed during the day (lights “on”), individual larval CoTS were, at times, seen swimming throughout the water column, but larvae tended to congregate in the upper third. Trichomes appeared to be evenly distributed throughout the water column, except in the high density treatment (100 trichomes ml^{-1}) where an estimated 20 to 50% of trichomes floated at the surface in most replicates. The distribution of larval CoTS and trichomes were not observed at night (lights “off”).

After 3 and 24 hours, 10 to 15 larvae from each replicate were examined using a fluorescence compound microscope (Olympus BX-53, XCite Series 120Q, MiChrome6 5Pro digital camera) under bright-field, unfiltered UV, and UV-chlorophyll filter (Olympus 19010-AT-FM 1-43/Chlorophyll Longpass) settings. When exposed to UV light, phytoplankton fluoresced red (Fig. 1). The proportion of larvae with material in the digestive tract was calculated for each replicate as the number of larvae with fluorescing material in their digestive tract out of the total number of larvae examined. Unfed larvae did not have fluorescing material in their digestive tract at any time. Data were statistically analyzed by ANOVA (Supplementary Text).

Larvae incorporate N fixed by *Trichodesmium*

Initial trials confirmed that the addition of ^{15}N - N_2 gas to sealed culture vessels resulted in the ^{15}N enrichment of *T. erythraeum* cells and the DOM exuded by these cells. To label *T. erythraeum* with ^{15}N , 50 ml of growth medium (modified YBC-II) was added to 50 ml of *T. erythraeum* culture in 100-ml serum vials. The headspace within each vial was degassed under vacuum, and ~20 ml of clean ^{15}N - N_2 (>98%; Cambridge Isotope Laboratories) was added. Vials remained sealed for ~72 hour, after which labeled *T. erythraeum* trichomes were separated via gentle filtration (0.45- μm polycarbonate). The labeled trichomes were washed three times with FSW and resuspended in fresh FSW. The filtrate containing ^{15}N -labeled DOM was kept for subsequent incubations. *T. erythraeum* trichomes from unlabeled cultures were prepared in the same way as ^{15}N -labeled trichomes.

CoTS larvae (8 dpf) were exposed to five larval feeding treatments: ^{15}N -labeled trichomes, ^{15}N -labeled DOM, the combination of ^{15}N -labeled DOM and ^{15}N -labeled trichomes, unlabeled trichomes, and FSW only (Unfed). Each treatment had five replicates, which consisted of a polypropylene vial (Techno Plas P8027UU) containing the larval feeding treatment and FSW to a total volume of 30 ml, and 50 CoTS larvae that had not been fed for 24 hours. Labeled or unlabeled trichomes were added to achieve a density of 100 trichomes ml^{-1} . Replicates in the labeled DOM and combined DOM-trichome treatments had 15 ml of DOM and 15 ml of FSW. The Unfed treatment had 30 ml of FSW only. The vials were sealed and opened every 12 hours for gas exchange. Larvae were exposed to the larval feeding treatments for 48 hours, after which larvae were moved to new vials and the seawater was exchanged for new FSW in all treatments via reverse filtration. The larvae were kept in clean FSW for 48 hours until their digestive tracts were empty when checked microscopically and then transferred via pipette to 2-ml Eppendorf tubes, rinsed three times in FSW, freeze dried, and stored at -20°C before ^{15}N compositional analysis.

Given that the small amount of N associated with the CoTS larvae, a ^{15}N analysis procedure previously used for determining the $\delta^{15}\text{N}$ of dissolved organic N was adopted (55). Briefly, larval samples were oxidized with $\text{K}_2\text{S}_2\text{O}_8$ and the $\delta^{15}\text{N}$ of the resulting NO_3^- was determined via the denitrifier protocol (56). The same procedure was used to measure the $\delta^{15}\text{N}$ of the DOM and the *T. erythraeum* cells (labeled and unlabeled). Data were statistically analyzed by ANOVA (Supplementary Text).

Larvae fed *Trichodesmium* develop normally

Actively swimming CoTS larvae (5 dpf) were placed into 40 1-liter rearing containers with FSW (26°C , 1- μm -filtered) at an initial density of 0.6 larvae ml^{-1} . Each container had gentle aeration from the base (31, 41, 42), which kept larvae and trichomes evenly mixed. FSW was exchanged daily, and containers exchanged every 5 days. Containers were fed once per day with one of the four diet treatments: *P. sulcata* (5×10^4 cells ml^{-1}), *T. erythraeum* (20 trichomes ml^{-1} ; mean trichome length = $357.4 \mu\text{m} \pm 29.9 \text{ SE}$, $n = 24$; mean number of cells per trichome = $51.9 \pm 5.8 \text{ SE}$, $n = 17$), *T. erythraeum* culture that had been filtered (0.45 μm) to remove trichomes, added at a volume equivalent to the amount of *T. erythraeum* culture added to achieve a density of 20 trichomes ml^{-1} (typically 10 to 25 ml per replicate), or were not fed ($n = 10$ rearing containers per treatment).

The density of larvae in each container was measured at 7 dpf and every 3 days thereafter until 25 dpf using previously described

methods (31, 41–43). Survival at each time point was calculated for each replicate using density values and reported as a percentage of the original density present at the beginning of the experiment. Development was monitored daily from 11 dpf and assessed following previously described methods in all replicates at 13 and 19 dpf when larvae in some treatments first reached the late brachiolaria stage indicative of morphological competence to settle (31, 41–43). At 10 and 16 dpf, ~20 larvae from each container were haphazardly collected, relaxed in 7% MgCl_2 and FSW, and then fixed in 15% formalin. Fixed larvae were photographed within 7 days, and the photographs were used to measure length, width, length of ciliated bands, and area of normal larvae following previously described methods (31, 41, 43). Data were statistically analyzed by ANOVA (Supplementary Text).

Supplementary Materials

This PDF file includes:

Supplementary Text
Figs. S1 to S4
Tables S1 and S2
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